

Structural Bioinformatics

Oral Exam Preparation

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1 SB01 – Introduction to Structural Bioinformatics

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1. Definition of bioinformatics
2. Structural bioinformatics
3. Central dogma of molecular biology
4. Protein structure and function
5. Structure prediction problem
6. Sequence–structure gap
7. Protein folding problem
8. AlphaFold and AI-based prediction
9. Course objectives
10. Computational tools and applications

1.1 What is structural bioinformatics?

Answer

Key points:

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Exam-length answer:

Follow-up questions / remarks

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1.2 How does structural bioinformatics differ from classical bioinformatics?

Answer

Key points:

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Exam-length answer:

Follow-up questions / remarks

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1.3 What is the sequence–structure gap?

Answer

Key points:

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Exam-length answer:

Follow-up questions / remarks

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1.4 Why is protein structure important for understanding function?

Answer

Key points:

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Exam-length answer:

Follow-up questions / remarks

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1.5 What is the protein folding problem?

Answer

Key points:

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Exam-length answer:

Follow-up questions / remarks

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1.6 Why was protein structure prediction historically difficult?

Answer

Key points:

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Exam-length answer:

Follow-up questions / remarks

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1.7 How did AlphaFold change the field?

Answer

Key points:

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Exam-length answer:

Follow-up questions / remarks

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1.8 What disciplines contribute to bioinformatics?

Answer

Key points:

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Exam-length answer:

Follow-up questions / remarks

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1.9 What are the main applications of structural bioinformatics?

Answer

Key points:

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Exam-length answer:

Follow-up questions / remarks

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1.10 What computational skills are important in this field?

Answer

Key points:

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Exam-length answer:

Follow-up questions / remarks

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2 SB02 – Chemical Bonds

Table of Contents

1. Matter and chemical transformations
2. Atomic theory
3. Atomic structure
4. Electrons and orbitals
5. Chemical bonding principles
6. Covalent bonds
7. Ionic interactions
8. Electronegativity
9. Molecular geometry
10. Non-covalent interactions in biomolecules

2.1 What is the difference between physical and chemical transformations?

Answer

Key points:

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Exam-length answer:

Follow-up questions / remarks

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2.2 Describe the structure of an atom.

Answer

Key points:

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Exam-length answer:

Follow-up questions / remarks

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2.3 What are orbitals and quantum numbers?

Answer

Key points:

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Exam-length answer:

Follow-up questions / remarks

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2.4 Explain the Aufbau principle.

Answer

Key points:

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Exam-length answer:

Follow-up questions / remarks

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2.5 What is electronegativity?

Answer

Key points:

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Exam-length answer:

Follow-up questions / remarks

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2.6 Compare covalent and ionic bonds.

Answer

Key points:

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Exam-length answer:

Follow-up questions / remarks

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2.7 What determines bond polarity?

Answer

Key points:

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Exam-length answer:

Follow-up questions / remarks

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2.8 Why are non-covalent interactions important in proteins?

Answer

Key points:

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Exam-length answer:

Follow-up questions / remarks

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2.9 What role do hydrogen bonds play in biology?

Answer

Key points:

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Exam-length answer:

Follow-up questions / remarks

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2.10 Why is carbon particularly suited for life?

Answer

Key points:

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Exam-length answer:

Follow-up questions / remarks

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3 SB03 – Amino Acids / Acid-Base Theory

Table of Contents

1. Arrhenius theory
2. Brønsted–Lowry theory
3. Acids and bases in aqueous solution
4. Strong vs weak acids
5. Acid dissociation constant
6. pKa
7. pH and pOH
8. Water self-ionization
9. Amino acid ionization
10. Buffer systems

3.1 Compare Arrhenius and Brønsted–Lowry theories.

Answer

Key points:

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Exam-length answer:

Follow-up questions / remarks

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3.2 What is a conjugate acid-base pair?

Answer

Key points:

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Exam-length answer:

Follow-up questions / remarks

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3.3 Define K_a and pK_a .

Answer

Key points:

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Exam-length answer:

Follow-up questions / remarks

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3.4 What is the relationship between pK_a and acid strength?

Answer

Key points:

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Exam-length answer:

Follow-up questions / remarks

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3.5 What is the significance of pH?

Answer

Key points:

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Exam-length answer:

Follow-up questions / remarks

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3.6 Why is water amphoteric?

Answer

Key points:

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Exam-length answer:

Follow-up questions / remarks

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3.7 What is a buffer?

Answer

Key points:

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Exam-length answer:

Follow-up questions / remarks

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3.8 Explain the Henderson–Hasselbalch equation.

Answer

Key points:

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Exam-length answer:

Follow-up questions / remarks

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3.9 How do amino acids behave at different pH values?

Answer

Key points:

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Exam-length answer:

Follow-up questions / remarks

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3.10 What is the isoelectric point?

Answer

Key points:

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Exam-length answer:

Follow-up questions / remarks

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4 SB04 – Protein Folding

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1. Protein folding problem
2. Native and denatured states
3. Levinthal's paradox
4. Secondary structure
5. Folding pathways
6. Energy landscape
7. Thermodynamics of folding
8. Folding kinetics
9. Chaperones
10. Misfolding and aggregation

4.1 What is Levinthal's paradox?

Answer

Key points:

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Exam-length answer:

Follow-up questions / remarks

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4.2 Why is protein folding not a random search?

Answer

Key points:

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Exam-length answer:

Follow-up questions / remarks

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4.3 What stabilizes the native state?

Answer

Key points:

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Exam-length answer:

Follow-up questions / remarks

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4.4 Explain the energy funnel model.

Answer

Key points:

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Exam-length answer:

Follow-up questions / remarks

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4.5 What are the main secondary structures?

Answer

Key points:

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Exam-length answer:

Follow-up questions / remarks

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4.6 What is the hydrophobic effect?

Answer

Key points:

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Exam-length answer:

Follow-up questions / remarks

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4.7 Difference between thermodynamics and kinetics of folding?

Answer

Key points:

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Exam-length answer:

Follow-up questions / remarks

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4.8 What are molecular chaperones?

Answer

Key points:

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Exam-length answer:

Follow-up questions / remarks

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4.9 Why do proteins misfold?

Answer

Key points:

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Exam-length answer:

Follow-up questions / remarks

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4.10 How is folding related to disease?

Answer

Key points:

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Exam-length answer:

Follow-up questions / remarks

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